Decision Trees
Intro

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Models we learned so far

Parametric vs. Non-parametric

Parameters vs. Hyperparameters

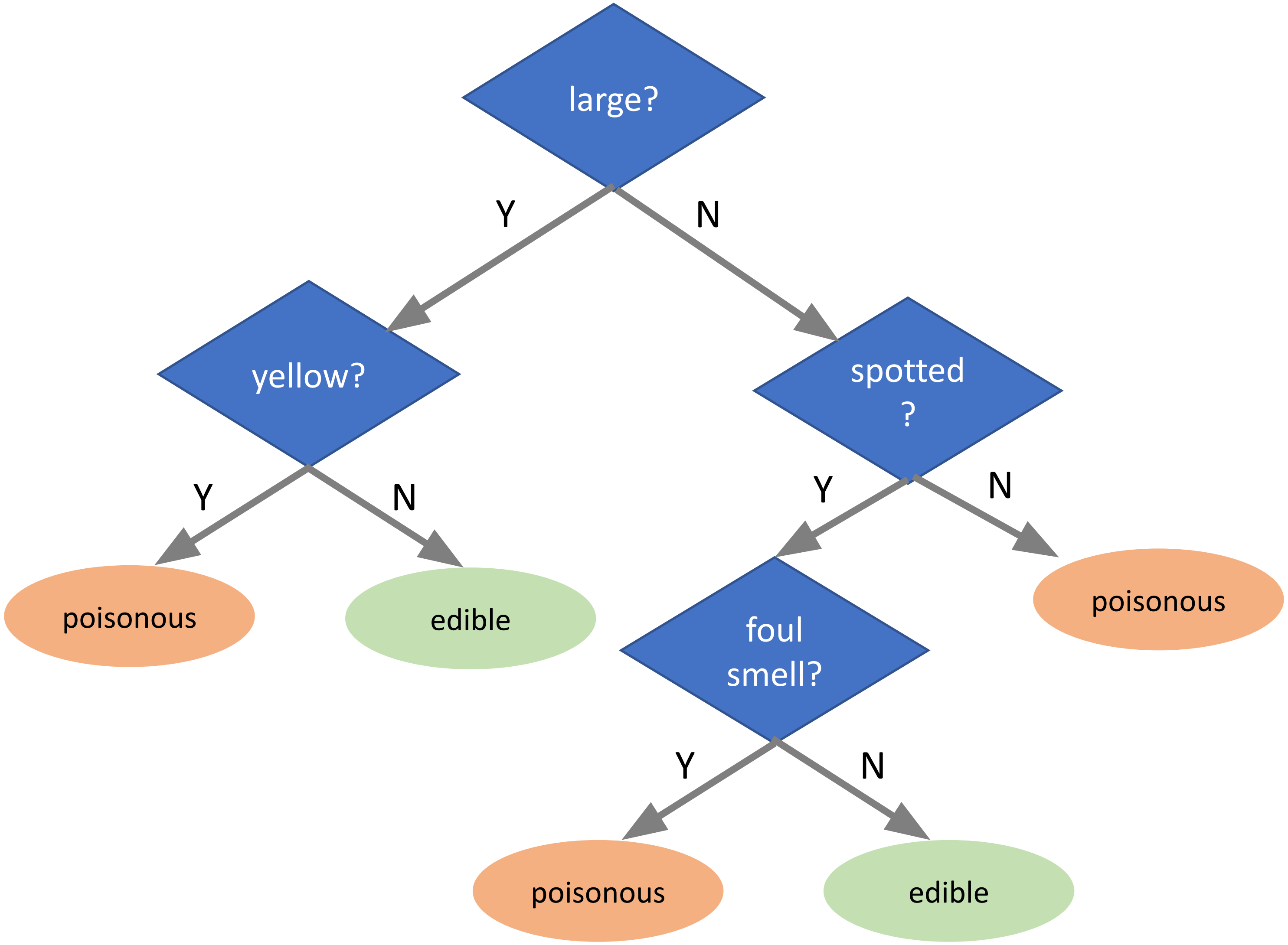
Linear Regression

Logistic Regression

kNN

Decision Tree

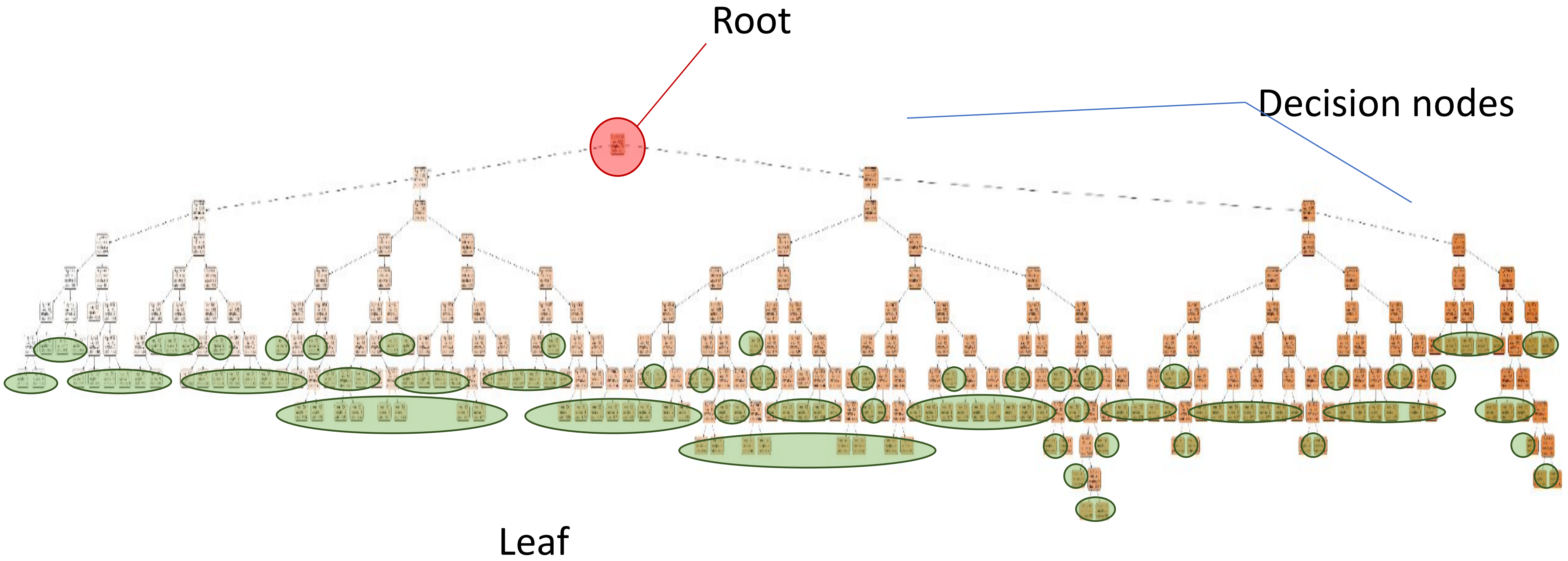
What is Decision Tree?

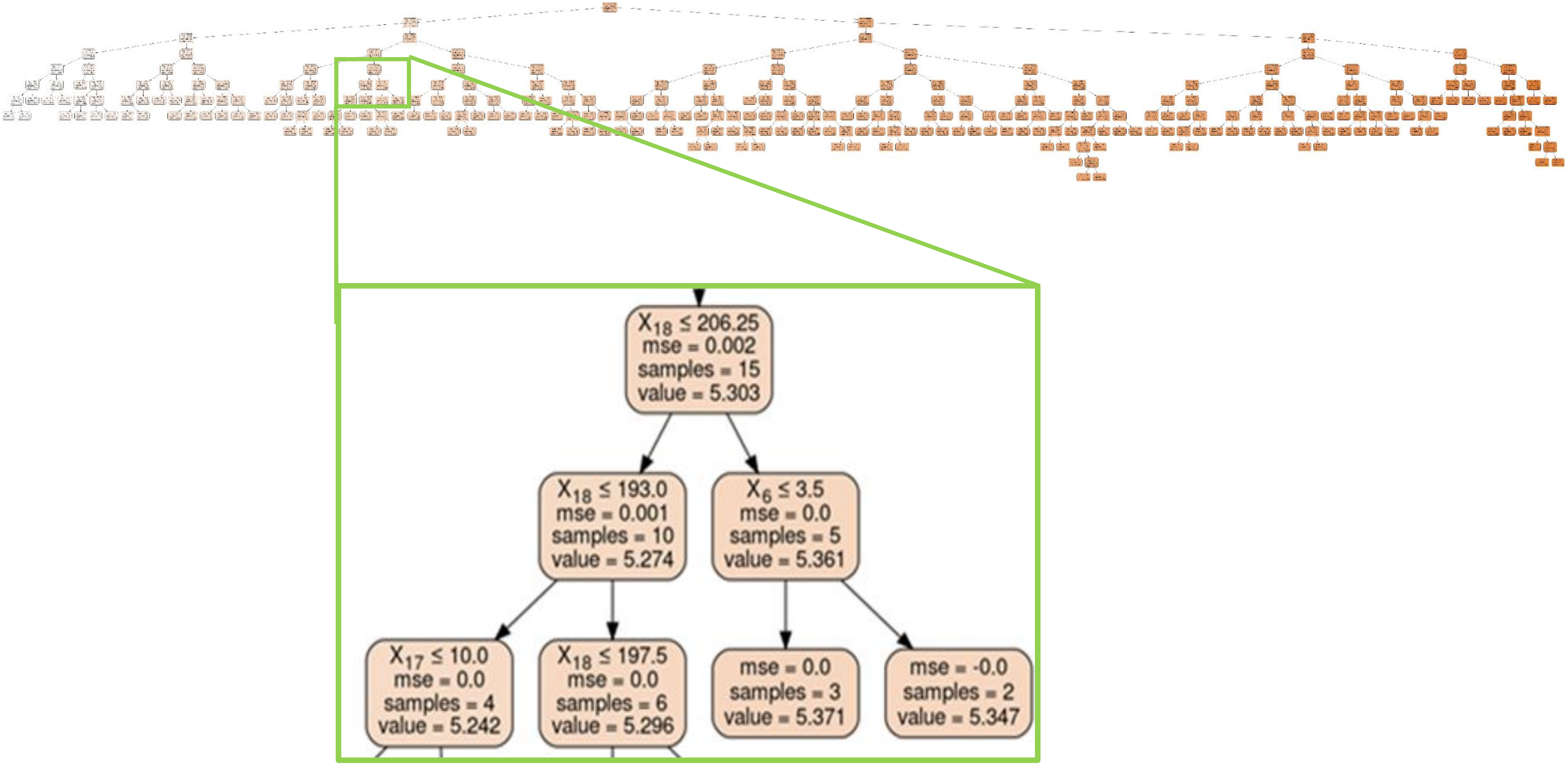



Caesar’s mushroom


Death Cap

Decision Tree Nodes




$X_{18} \leq 206.25$
mse = 0.002
samples = 15
value = 5.303
$X_{18} \leq 193.0$
mse = 0.001
samples = 10
value = 5.274
$X_6 \leq 3.5$
mse = 0.0
samples = 5
value = 5.361
$X_{17} \leq 10.0$
mse = 0.0
samples = 4
value = 5.242
$X_{18} \leq 197.5$
mse = 0.0
samples = 6
value = 5.296
mse = 0.0
samples = 3
value = 5.371
mse = -0.0
samples = 2
value = 5.347

How models "Learn" to make a Decision

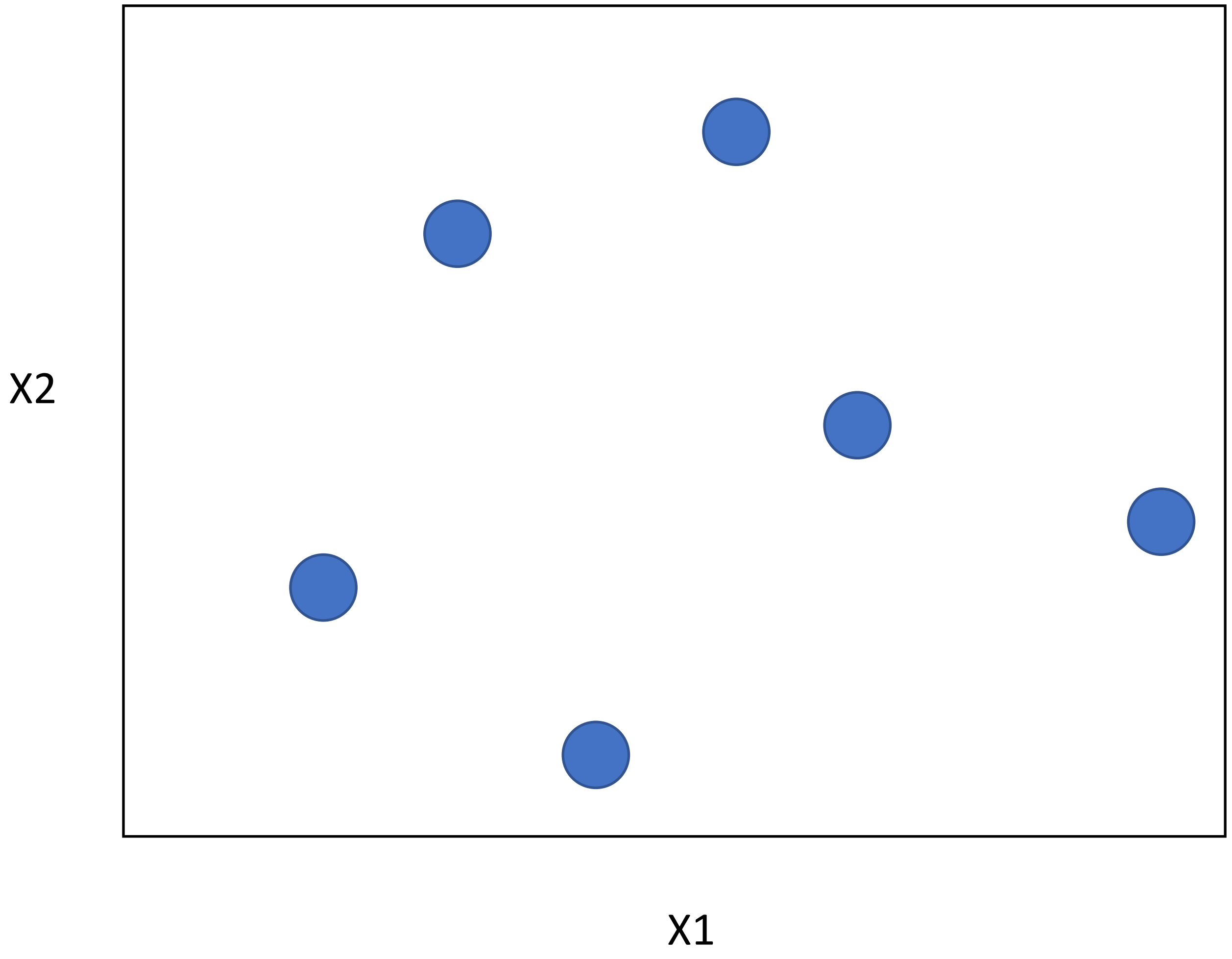
Linear Regression	Minimize MSE
Logistic Regression	Minimize Cross Entropy
kNN	No optimization, but uses distance
Decision Tree	Split to minimize MSE for Regression task and minimize Cross Entropy or Gini for Classification task

Decision Tree Regressor

The goal is to find boxes $R_1 \sim R_J$ such that

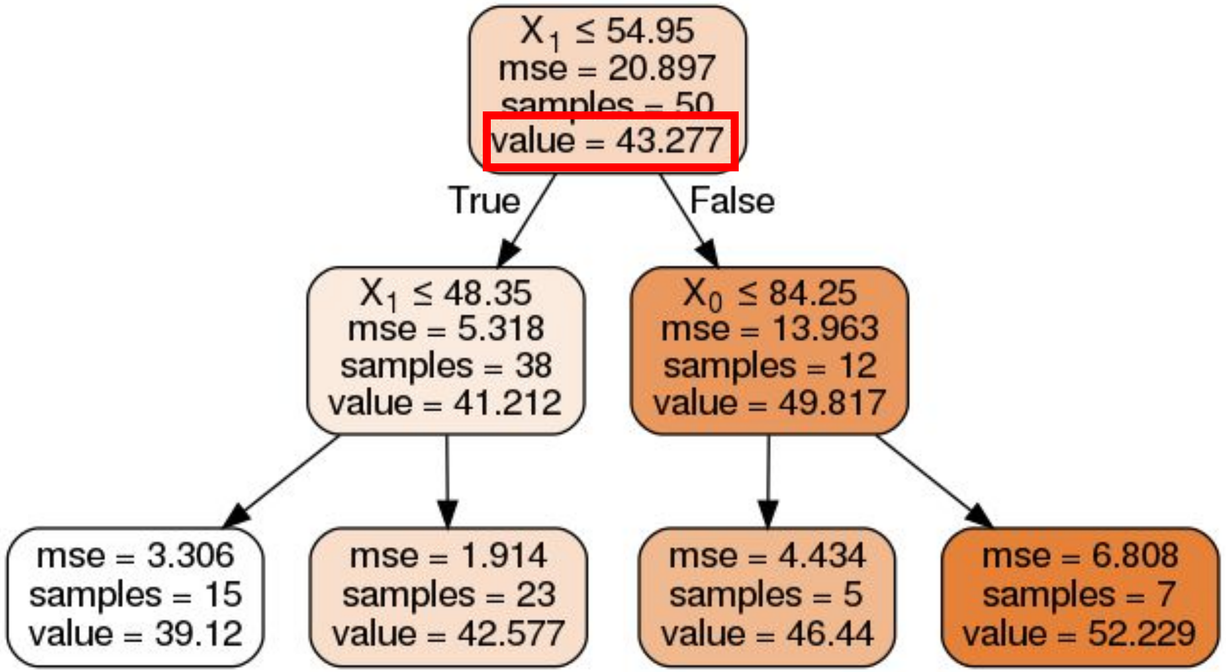
$$\sum_{j=1}^{J}\sum_{i\in R_j}(y_i - \hat{y}_{R_j})^2$$ is minimized.

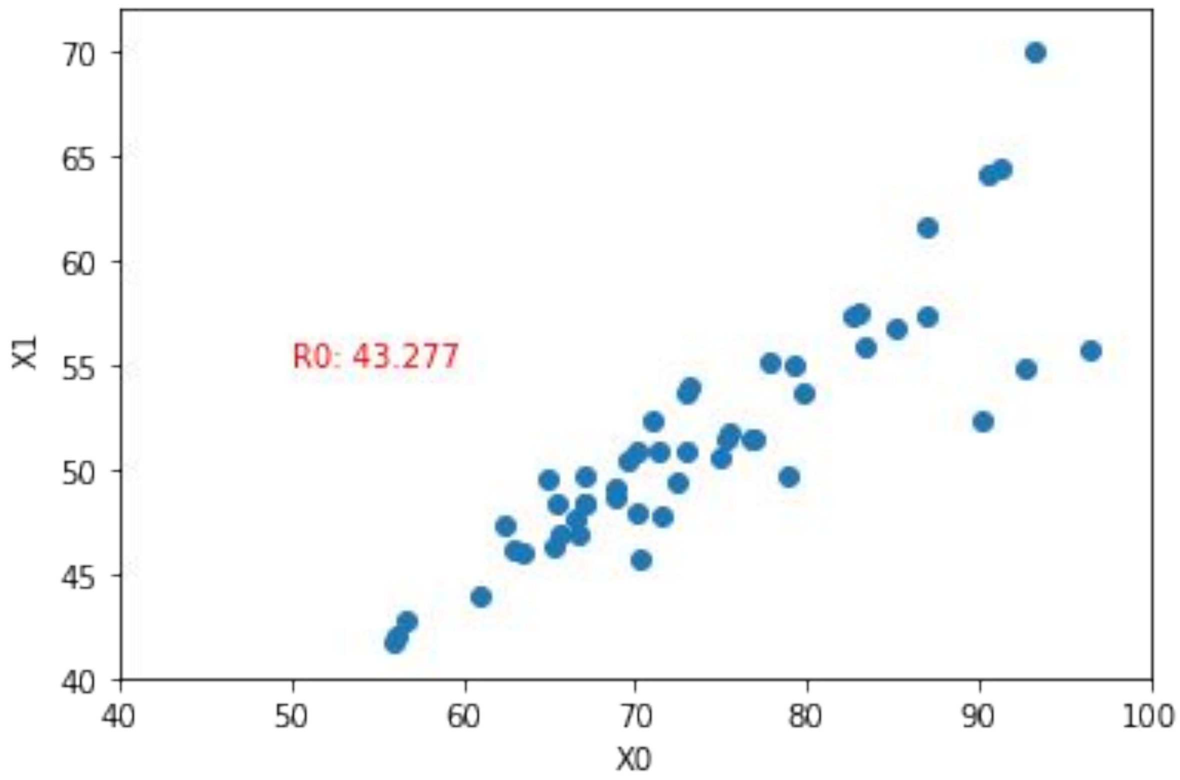
the mean of the data in the box



Decision Tree Regressor

Faculty Salary Dataset
4 features
50 samples





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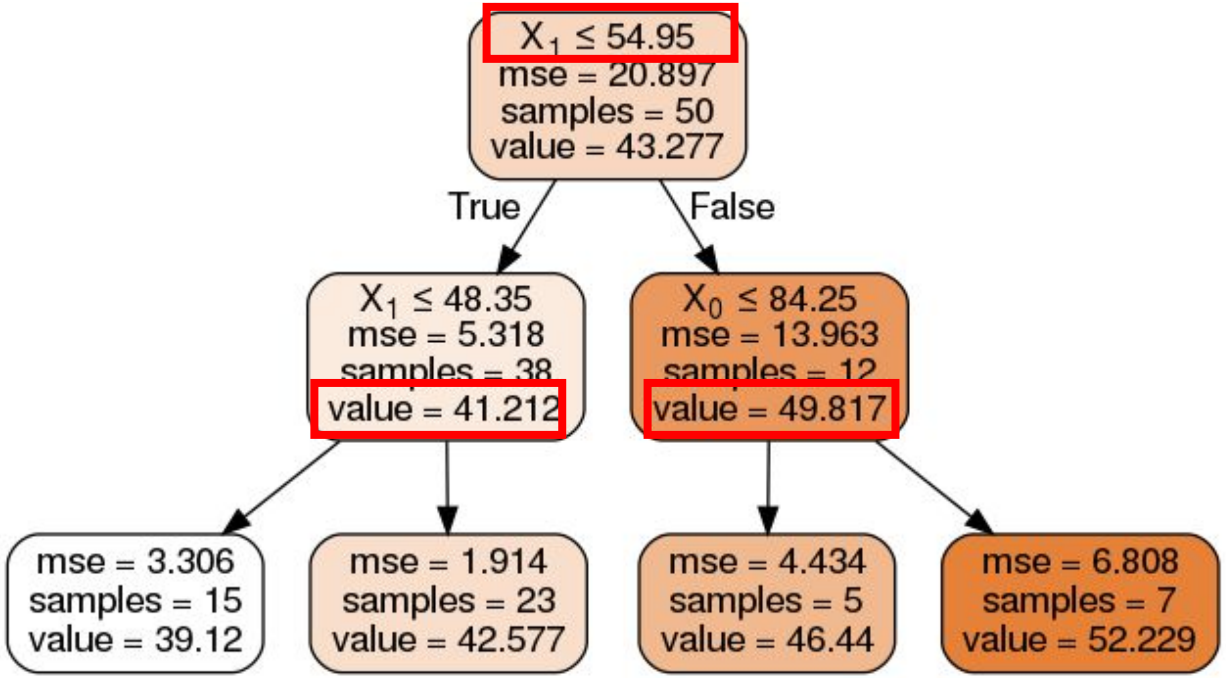

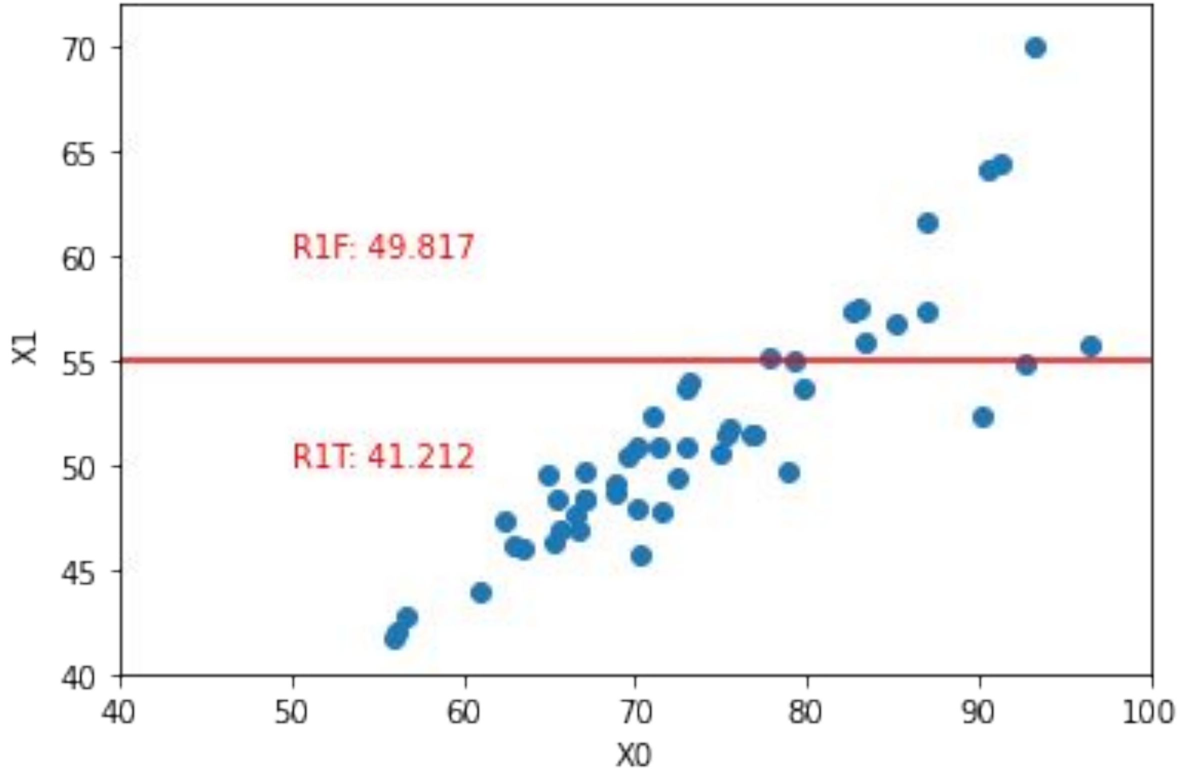

Decision Tree Regressor

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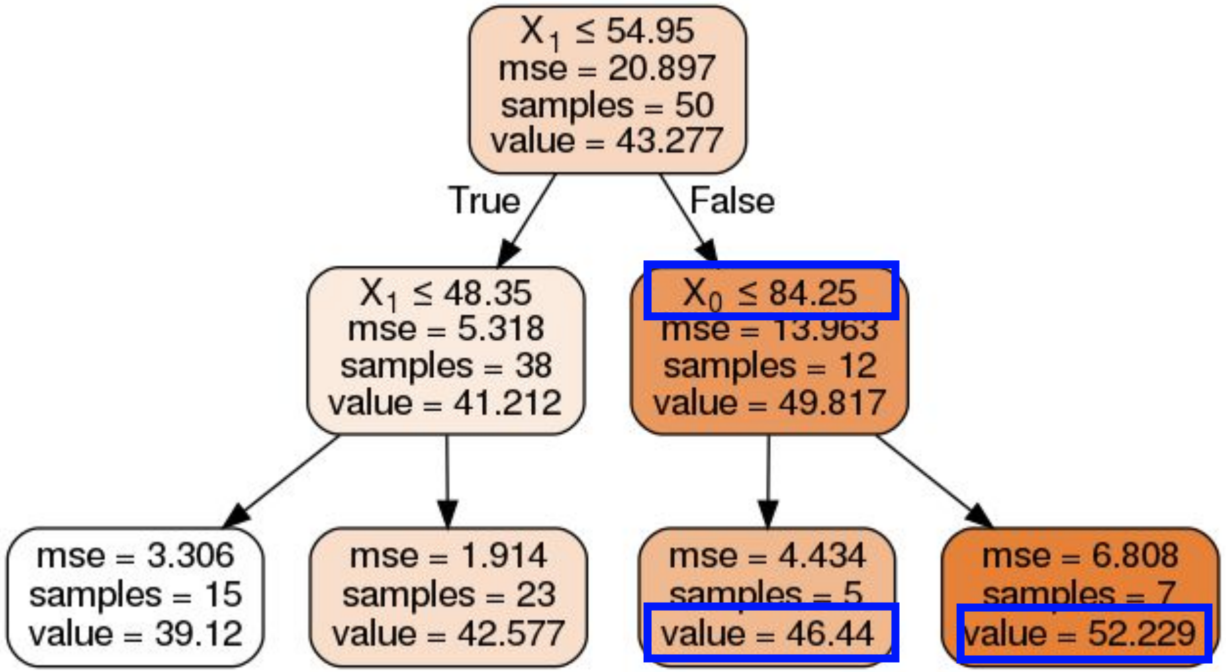

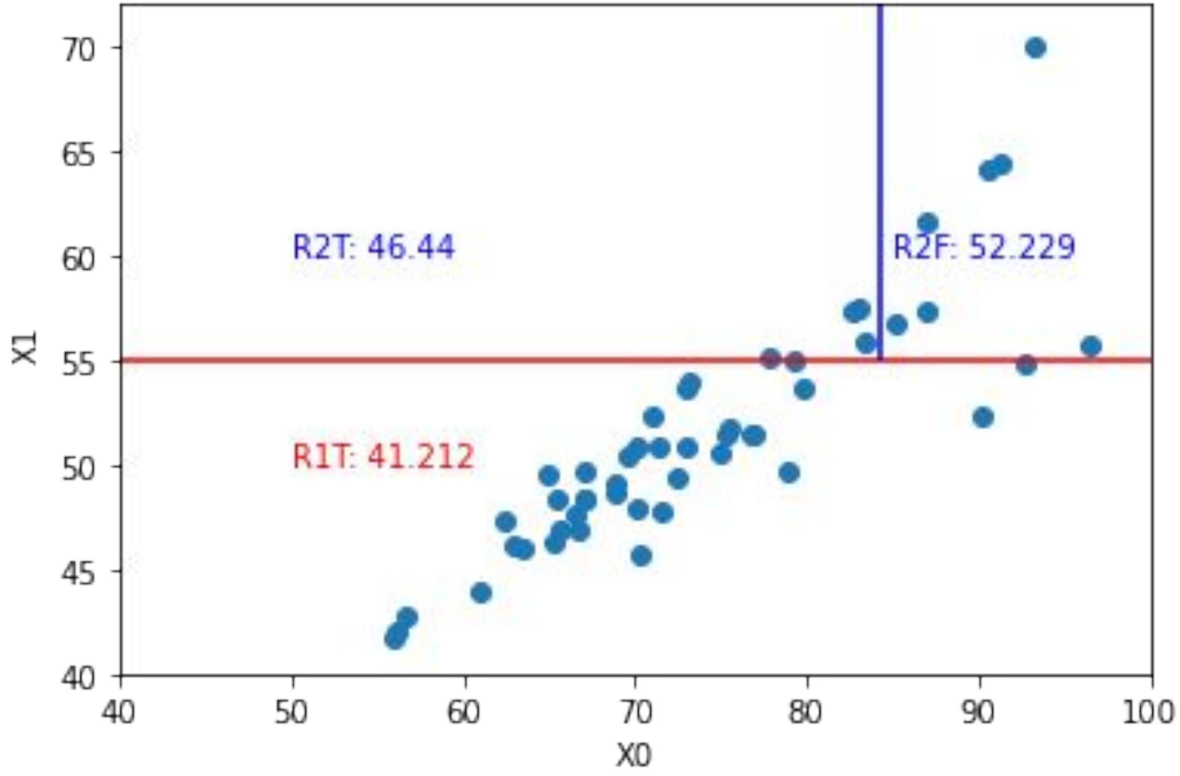

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